

# Validation Note

## 1. Frozen data and chronology

156 consecutive Monday-start weeks: 02 Oct 2023 to 21 Sep 2026. First 144 weeks train; final 12 weeks hold out. Fit cutoff is the week starting 29 Jun 2026, ending 05 Jul. Every candidate forecasts all 12 held-out weeks from the same origin after 05 Jul. Weekly demand is recorded retail sales in five-kilogram packs. No rows were dropped for a better score.

Frozen CSV SHA-256:  
b94ac87db74da82ec4360aef446667e8a1478c87497c226f32728fae85d79a1d

The dataset was frozen before model comparison. It was not regenerated during completion. The workbook retains the data dictionary, assumptions, exact Section 4.2 prompt and fixed seed 3532026.

## 2. Leakage and model construction

Naive repeats the final training value; seasonal naive repeats the corresponding training week 52 weeks earlier. SMA4 appends predicted values. SES uses fixed alpha 0.3 and 0.6. Both Holt variants use a disclosed training-only 160-candidate grid and fixed initialization. Event adjustment uses only training OLS and the known calendar, with zero event flags in this holdout. Neither realized outlets nor stockout tags enter the forecast function.

The notebook reruns forecasts while held-out actuals are perturbed and checks that forecasts are identical. It also checks SMA’s first two recursion steps, naive and seasonal indices, date ordering and the 144/12 partition. Nonzero errors rule out suspiciously perfect accuracy. Historical\_Forecast is a separate series of weekly planning forecasts and is not ranked as a common-origin competitor.

## 3. Holdout comparison

| Model                      | MAD   | MAPE   | RMSE  | WAPE   | Bias   | TS     |
|----------------------------|-------|--------|-------|--------|--------|--------|
| SMA (4 weeks, recursive)   | 553.6 | 10.51% | 752.7 | 10.26% | -125.5 | -2.72  |
| Event-adjusted damped Holt | 638.6 | 12.38% | 779.6 | 11.83% | -295.7 | -5.56  |
| SES (alpha 0.6)            | 641.1 | 12.44% | 781.1 | 11.88% | -299.5 | -5.60  |
| SES (alpha 0.3)            | 648.0 | 12.58% | 785.1 | 12.01% | -309.8 | -5.74  |
| Seasonal naive (52 weeks)  | 685.2 | 13.38% | 894.5 | 12.69% | -664.2 | -11.63 |
| Damped Holt                | 731.7 | 14.32% | 842.5 | 13.55% | -435.3 | -7.14  |
| Naive                      | 746.8 | 14.64% | 854.5 | 13.84% | -458.0 | -7.36  |

SMA4 is the provisional choice because it has lowest WAPE and RMSE, sensible level behavior and a moderate negative bias. Event-adjusted Holt and SES 0.6 are nearly tied. A single short holdout supports comparison, not a guarantee of future performance. Model selection uses this same holdout; there is no second untouched test set.

## 4. Independent metric arithmetic

Use Error = Actual minus Forecast. Positive means underforecasting, negative means overforecasting. Retain full precision until display. All metrics use the same 12 weeks.

| Metric          | Calculation / result                                                | Managerial meaning                                                                                               |
|-----------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| MAD / MAE       | $6,643.5398 / 12 = 553.6283$ packs                                  | Typical miss is about 554 packs, or 2.77 tonnes/week.                                                            |
| MAPE            | $\text{Mean}(\text{abs}(\text{error}) / \text{actual}) = 10.5123\%$ | Average weekly percentage error. No zero actuals occur; zeros or near-zero demand make it undefined or unstable. |
| RMSE            | $\text{sqrt}(6,798,327.5949 / 12) = 752.6801$ packs                 | Large misses receive more weight.                                                                                |
| WAPE            | $6,643.5398 / 64,776 = 10.2562\%$                                   | Total absolute error as a share of actual volume, not a service-level promise.                                   |
| Bias / MFE      | $-1,505.8066 / 12 = -125.4839$ packs                                | Average overforecast is about 125 packs/week.                                                                    |
| RSFE            | -1,505.8066 packs                                                   | Signed error accumulated over this window.                                                                       |
| Tracking Signal | $-1,505.8066 / 553.6283 = -2.7199$                                  | Negative means accumulated overforecasting. The $\pm 4$ band is an assumed review trigger, not a Matco policy.   |

Three metrics are independently implemented as Excel formulas in Dataset → Metric Verification: MAD, RMSE and WAPE. Those formulas recursively build SMA4 from training-only cells, then score actuals. They agree with separate notebook arithmetic. Automated verification is recorded; no student hand calculation is falsely claimed.

## 5. The five largest selected-model misses

| Week       | Actual | Forecast | Error A-F | Diagnosis                                                                                                                                   |
|------------|--------|----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-07-13 | 6,997  | 5,436.7  | 1,560.3   | Promotion week: a level-only forecast cannot anticipate planned uplift; synthetic noise may also contribute.                                |
| 2026-08-24 | 4,083  | 5,538.7  | -1,455.7  | Transport-disruption/stockout scenario depresses recorded sales; forecast exceeds constrained sales. Unmet customer demand is not measured. |
| 2026-08-03 | 6,448  | 5,507.3  | 940.7     | Promotion week: a level-only forecast cannot anticipate planned uplift; synthetic noise may also contribute.                                |
| 2026-09-07 | 4,766  | 5,534.7  | -768.7    | Ordinary week: unexplained variation and possible level/coverage mismatch; no promotion, event or stockout tag supports a specific cause.   |
| 2026-08-10 | 4,787  | 5,524.4  | -737.4    | Ordinary week: unexplained variation and possible level/coverage mismatch; no promotion, event or stockout tag supports a specific cause.   |

Tags support plausible explanations rather than proven causes. Stockout-week sales are availability-constrained and do not reveal lost customer demand. The normal tagged weeks still contain unexplained variation.

## 6. Regression audit and uncertainty

Training-only OLS uses Price, Promo\_Flag, Event\_Flag and Distribution\_Points (n=144). Independent centered/scaled QR estimates match the coefficients.  $R^2 = 0.611$ , adjusted  $R^2 = 0.600$ . HAC uses four lags; robust p-values are approximate. Promotion and calendar associations remain positive in the training stockout-exclusion sensitivity. Price remains uncertain. VIF about 2.9 for price/coverage and a common time trend limit their separation. Omitted promo depth, competitor effects, store mix and constraints can bias interpretation.

The forward SMA refit uses all available history after preserving original holdout scores. The 81 training rolling origins create a descriptive 10th-90th pointwise residual envelope. Origins overlap; this is not a calibrated confidence interval. It covers 10 of 12 holdout actuals retrospectively (83.3%) without retuning. The holdout contains no calendar-event weeks, so it does not test Ramadan/Eid adjustment.

## 7. Decision stress test and conservation audit

Assume 6,000 regular packs/week, 7,000 with reserved option, 2,500 opening/target buffer, one-week replenishment lead, first two weeks firm and cost PKR 2,400/pack. After week two, receipts use last week's demand and opening inventory. Shortages are unfilled volume; no backorders carry forward. Week-three delivery interruption sets receipts to zero and bounds later recovery by weekly capacity.

| Scenario                         | Demand packs | Regular shortage | Option shortage | Peak stock packs |
|----------------------------------|--------------|------------------|-----------------|------------------|
| Base                             | 63,478       | 0                | 0               | 2,528            |
| Downside 10%                     | 57,130       | 0                | 0               | 3,551            |
| Downside 20%                     | 50,782       | 0                | 0               | 4,601            |
| Upside 10%                       | 69,826       | 0                | 0               | 2,505            |
| Upside 20%                       | 76,174       | 3,171            | 0               | 2,500            |
| Promotion shock +30% (weeks 3-4) | 66,669       | 114              | 0               | 2,507            |
| Delivery interruption (week 3)   | 63,478       | 2,824            | 2,824           | 2,507            |

Every weekly scenario satisfies Opening + Receipts + Unfilled = Demand + Ending, with nonnegative receipts, stock and shortage. The downside tests quantify temporary stock exposure; upside and interruption tests quantify unfilled volume. Cost values use an explicitly assumed cost rather than the retail price.

## 8. What could reject or change the recommendation?

Revise the plan if actual capacity, stock, lead time or costs contradict assumptions; approved campaigns change demand; a fresh holdout reverses model selection; or stockouts conceal unmet demand. Verify the capacity option and backup route before a real commitment.

## 9. Verification status and student review

Checks executed: source hash, chronology, partition, forecast recursions, all metrics, regression QR and inventory balances. Excel reproduces three metrics. Student manual checking, comprehension and LMS/GitHub submission remain the student's responsibilities.